

What Survives the Relaxation

Four boundaries, from amnesia to agency

Reading and note-making task • MYP • Maths, Physics & Biology

Part A Before you read: make a prediction

Do not read the essay yet. Answer from what you already think.

A soap bubble always ends up as a sphere, no matter who blew it or how. A protein is a long chain of chemicals that folds up into a shape, and every different protein folds into a *different* shape.

A1. Both of these happen by themselves, with no one shaping them. So why does one always end the same, and the other always end differently? Write your best guess.

A2. Which of these do you think holds more *information* about how it was made: a bubble, or a folded protein? Circle one and give one reason.

BUBBLE

PROTEIN

Keep this page. You will come back to it at the end and mark your own prediction.

Glossary strip use while reading

free energy	the part of a system's energy that is actually available to do work
dissipative	losing energy to the surroundings as it changes; it cannot go back up
relaxation	a system settling downhill into a low-energy shape and stopping there
partition	dividing one space into many cells, like a honeycomb or a foam
protein folding	a chain of amino acids collapsing into its working 3D shape
residue	one amino-acid unit in that chain
binding pocket	the cleft in a folded protein that one particular molecule fits into
flux	a continuous flow of energy through something, keeping it going

Part B Cornell notes: the four boundaries

Fill the **Notes** column as you read. Some cues are given; write your own in the blank cue boxes.

Cue 1. THE BUBBLE Why is every bubble of the same size identical? What does “a boundary with no memory” mean?	Notes
Cue 2. THE FOAM In 2D the winner is the hexagon. What happens in 3D? Where does irregularity first appear?	Notes
Cue 3. THE HONEYCOMB The bees don’t calculate the hexagon. What does? What one fact about the larva survives in the wax?	Notes
Cue 4. THE PROTEIN Same downhill process as the bubble. Why a different outcome? <i>Your own cue:</i> _____	Notes

Cue	Notes
5. HISTORY IN THE CHAIN Why isn't a protein's shape purely designed for its present job? <i>Your own cue:</i> _____	

Cue	Notes
6. THE MACHINE What is the difference between a shape that has finished falling and one that is being driven? <i>Your own cue:</i> _____	

Summary box. In no more than **40 words**, and without using the words *bubble*, *bee* or *protein*, say what the whole essay argues.

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Part C Cloze: the argument in one paragraph

Fill each gap. Every answer is one or two words. Six of the eight are in the glossary strip; two are not.

Every one of these boundaries is made by the same physical process: something settles _____ into the lowest shape available to it and then stops. The physics is the same every time; what changes from case to case is where the _____ sits.

In the bubble, it sits in the starting shape, so the fall _____ it, and the result is a sphere that tells you nothing about how it was made.

In the honeycomb, the hexagon comes from the wax itself, but the _____ of each cell is set by the size of the _____ that has to fit inside, so exactly one fact about the occupant survives.

In the protein, the order of the _____ in the chain decides which shapes are low and which are high, so the same downhill fall *reveals* the information instead of destroying it. That is why its surface is _____ rather than repeating.

The last step is different in kind. A molecular machine does not just sit at the bottom of its fall; it burns _____ every cycle, and stops working the moment the supply stops.

Part D Understanding and reflection

1. **Sorting task.** Put each item in the right column. *Snowflake, kinesin motor protein, raindrop, collagen fibre, sand dune ripple, enzyme active site.*

Shape from the medium alone	Shape from an inherited sequence	Needs a constant energy supply

One of these six was harder to place than the others. Which, and why?

2. **The claim you must argue with.** The essay says irregularity is not a failure to be neat; it is what the best answer looks like when a surface has a different job at every point. Give one example *from outside this essay*, from anything at all, where an irregular design is better than a regular one, and say what makes it so.

3. **Testing the limit.** A flame is fed energy continuously and holds a definite shape, but the essay says it is not like a molecular machine. Explain the difference in your own words.

4. **Back to your prediction.** Reread your answer to A1. Mark it yourself: what did you get right, and what would you now change?

5. **One honest question.** Write down one thing in this essay you still do not understand. Be specific; "all of it" is not a question.

Teacher key and notes

Part C cloze. downhill; information; destroys; width; larva; residues; irregular; free energy (accept “fuel” or “ATP”).

Part D1 sorting. Medium alone: snowflake, raindrop, sand dune ripple. Inherited sequence: collagen fibre, enzyme active site. Constant energy: kinesin. The intended difficulty is the snowflake, which is irregular in outline yet holds no inherited information, so it separates “irregular” from “carries history” and stops students equating the two. The sand dune ripple is a fair second answer, since it is driven by wind while it forms and so sits awkwardly between columns one and three; accept either if the reasoning is sound.

Part D3. The expected distinction is that a flame consumes energy but leaves nothing behind that a later flame inherits, whereas a molecular machine belongs to a lineage that retains and passes on its structure. Students who say only “a flame is not alive” have restated the question; push for what living things do that fire does not.

Pitching note. The source essay is written well above Year 9 reading level. It works if Parts A and the glossary strip are done first and the essay is read in the six chunks the cue column already divides it into, one boundary at a time, rather than end to end. Sections 5 and 6 are the hardest; they can be teacher-read aloud with the notes column filled together.

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